

Assignment 2 - CSE340

1. Convert this C code to the corresponding RISC V format instructions.

```
long long int max(long long int a, long long int b){  
    if(a > b)  
        return a;  
    else  
        return b;  
}
```

2. Write a function to find if a number is even or odd in RISC-V assembly language.
3. Write RISC V code for the following code. Consider the base address of A is in x10 and X and Y are in registers x8 and x9 respectively. Make sure you make optimum utilization of the registers.

A[3] = 12*X - 7*A[5] + 33*Y - 70;

In this code, if there are any I format instructions, convert those to its corresponding machine code.

4. Construct the equivalent RISC-V code of the following C code. Once you have the RISC-V code, identify the type of each instruction and encode them accordingly.

A[7] = A[2] + A[B[8]] + 10;

B[i] = A[3] - 8;

Base addresses of array A and B are in register X20 and X21 and i is in register X22.

5. A=[1,2,3,4,5], B = [6,7,8, 9,0]

RISC V code:

slli x30, x5,3

f = x5 = 2

add x30,x10,x30

g = x6 = 1

slli x31, x6, 3

h = x7

add x31, x11,x31

i = x8

ld x5,0(x30)(1)

j = x9

addi x12,x30,8

ld x30,0(x12)(2)

base address of A = x10

add x30,x30,x5(3)

base address of B = x11

sd x30,0(x31)

Convert this assembly code to any high level language of your choice. Find the values of the register x5 in label (1), x30 in label (2) and label (3).